

Movies Dataset Analysis - Detailed Summary Report

1. Libraries Imported: - NumPy and Pandas for numerical and data manipulation tasks. - Seaborn and Matplotlib for statistical data visualization.

2. Dataset Loading: - The dataset 'mymoviedb.csv' was read into a Pandas DataFrame. - This dataset contains information about movies, including release dates, language, votes, and more.

3. Initial Exploration: - `df.head()`: Displayed the first few rows to get a quick view of the dataset. - `df.info()`: Provided column details, data types, and null value counts. - `df.duplicated().sum()`: Checked for duplicate rows to maintain data quality. - `df.describe()`: Generated descriptive statistics such as mean, median, min, max, and standard deviation.

4. Data Preprocessing: a) Date Handling: - Converted 'Release_Date' column into datetime format. - Extracted only the year component for uniform analysis. b) Dropping Columns: - Removed columns 'Overview', 'Original_Language', and 'Poster_Url' as they were not relevant for the analysis. c) Vote Average Categorization: - The 'Vote_Average' column was categorized into bins (e.g., low, medium, high) for easier interpretation.

5. Data Cleaning Outcomes: - The dataset was standardized, redundant information was dropped, and structured for deeper analysis.

6. Visualization and Analysis: - Various visualizations were prepared to analyze distributions, trends, and relationships within the dataset. - Visual analysis using Seaborn and Matplotlib enhanced understanding of vote averages, release trends, and categorical insights.

Conclusion: This notebook demonstrates a systematic approach to handling raw movie data. It involves cleaning, preprocessing, and preparing the dataset for effective analysis. With consistent formatting and relevant features retained, the dataset is well-prepared for advanced analytics, such as trend analysis, recommendation systems, and predictive modeling.